Inequalities and optimisation

L3-AL2 Level 3 Senior · Algebra

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How to find and prove largest and smallest values using completing the square, AM-GM, the chain of means and extreme-case reasoning.
By the end you can find maxima and minima by completing the square and by AM-GM with well-chosen splits; state, prove and use the QM-AM-GM-HM inequalities for two numbers; test extreme cases and check that a bound is actually attained.
Prerequisites: Factorising, substitution and systems (L2-AL1), Sequences, sums and averages (L2-AL2), Polynomials and Vieta's formulas (L3-AL1). **Contests:** UKMT SMC, Senior Kangaroo, Maclaurin, AMC 12, AIME. **Time:** about 150 minutes.

1 Lesson

Consider the smallest value of $x + \frac{9}{x}$ for positive x. Trying a few values, $x = 1$ gives 10, $x = 2$ gives 6.5, $x = 3$ gives 6 and $x = 4$ gives 6.25. This suggests the answer 6, at $x = 3$, but a table of values proves nothing. A one-line proof is

$$x + \frac{9}{x} - 6 = \frac{x^2 - 6x + 9}{x} = \frac{(x-3)^2}{x} \geq 0.$$

So $x + \frac{9}{x} \geq 6$ for every positive x, with equality only when $x = 3$. Every optimisation answer needs both parts seen here: a *bound* (the expression is never below 6) and an *example* showing that the bound is achieved (at $x = 3$). Without the example the bound may not be the answer, and without the bound the example is only a guess.

1.1 Squares are never negative

Almost everything in this unit follows from the fact that a real square is never negative. The most direct use is completing the square. For $a > 0$,

$$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a},$$

so the minimum is $c - \frac{b^2}{4a}$, reached at $x = -\frac{b}{2a}$. With several variables, complete the square one variable at a time, treating the others as constants, and use up the cross terms first.

Example 1.
Find the minimum value of $x^2 + 2xy + 3y^2 - 4y + 5$ over all real x and y.

Solution.
A tempting idea is to minimise in each variable separately, for instance by setting $x = 0$ because $x^2 \geq 0$. That is wrong, because the term $2xy$ can be negative, so a non-zero x might help. So the cross term must be dealt with first.

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Treat the expression as a quadratic in x with y fixed: $x^2 + 2xy = (x+y)^2 - y^2$. So the expression equals $(x+y)^2 + 2y^2 - 4y + 5$, and completing the square in y gives

$$(x+y)^2 + 2(y-1)^2 + 3.$$

Both squares are non-negative, so the expression is at least 3. Equality needs $y = 1$ and $x = -y = -1$, and indeed $1 - 2 + 3 - 4 + 5 = 3$. The minimum is 3.

1.2 AM-GM for two numbers

AM-GM.
For non-negative a and b, $\frac{a+b}{2} \geq \sqrt{ab}$, with equality if and only if $a = b$.

Proof. $(\sqrt{a} - \sqrt{b})^2 \geq 0$ expands to $a + b - 2\sqrt{ab} \geq 0$, and equality holds exactly when $\sqrt{a} = \sqrt{b}$. The opening problem is a case of AM-GM: $x + \frac{9}{x} \geq 2\sqrt{x \cdot \frac{9}{x}} = 6$. This works because the product $x \cdot \frac{9}{x}$ is constant. AM-GM gives a constant lower bound for a sum when the product is fixed, and a constant upper bound for a product when the sum is fixed. Identifying which quantity is fixed is usually the hard step.

1.3 More terms, and splitting

AM-GM for n numbers.
For non-negative $a_1, \ldots, a_n$,

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \cdots a_n},$$

with equality if and only if all the a_i are equal.

Proof. For four numbers, use the two-number version twice:

$$\frac{a+b+c+d}{4} = \frac{1}{2}\left(\frac{a+b}{2} + \frac{c+d}{2}\right) \geq \frac{\sqrt{ab} + \sqrt{cd}}{2} \geq \sqrt[4]{abcd}.$$

The same doubling gives 8, 16, and so on. For three numbers, let $m = \frac{a+b+c}{3}$ and apply the four-number version to a, b, c, m. Their mean is $\frac{3m+m}{4} = m$, so $m \geq \sqrt[4]{abcm}$, which gives $m^4 \geq abcm$ and so $m^3 \geq abc$ (if $m = 0$ everything is 0 anyway). The same padding idea handles every n, and following the equality cases through each step shows that equality needs all the numbers equal.

Example 2.
For positive x, find the minimum value of $2x + \frac{27}{x^2}$.

Solution.
Two-term AM-GM gives $2x + \frac{27}{x^2} \geq 2\sqrt{\frac{54}{x}}$, but the right-hand side depends on x, so it is not a bound. We need pieces whose product is constant.
Here x appears to the power 1 in the first term and -2 in the second. Splitting $2x$ into two equal pieces $x + x$ fixes this, because $x \cdot x \cdot \frac{27}{x^2} = 27$. Three-term AM-GM now gives

$$x + x + \frac{27}{x^2} \geq 3\sqrt[3]{27} = 9.$$

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Equality needs $x = \frac{27}{x^2}$, that is $x^3 = 27$, so $x = 3$. Check: $6 + 3 = 9$. The minimum is 9.
In general, split terms into equal pieces so that the product of all the pieces is constant, and make sure the equality case can actually happen.

Example 3.
A cuboid has a square base. The total length of its twelve edges is 48 cm. Find the largest possible volume.

Solution.
Let the base have side x and the height be h. The edges give $8x + 4h = 48$, so $2x + h = 12$, and the volume is $V = x^2h$. Writing $V = x^2(12 - 2x)$ and trying values gives only a guess. Note instead that we have a fixed sum and want to bound a product, which is the second form of AM-GM.
The product is $x \cdot x \cdot h$, and the fixed sum $2x + h$ is $x + x + h$, the sum of the same three factors. So

$$4 = \frac{x + x + h}{3} \geq \sqrt[3]{x^2h} = \sqrt[3]{V},$$

giving $V \leq 64$. Equality needs $x = h$, and with $2x + h = 12$ that means $x = h = 4$, a cube. The largest volume is 64 cm^3.
If the fixed sum had been $x + h$ instead, you would split it as $\frac{x}{2} + \frac{x}{2} + h$, so that the product of the pieces is $\frac{V}{4}$, still a constant multiple of V.

1.4 The chain of means

QM-AM-GM-HM.
For positive a and b,

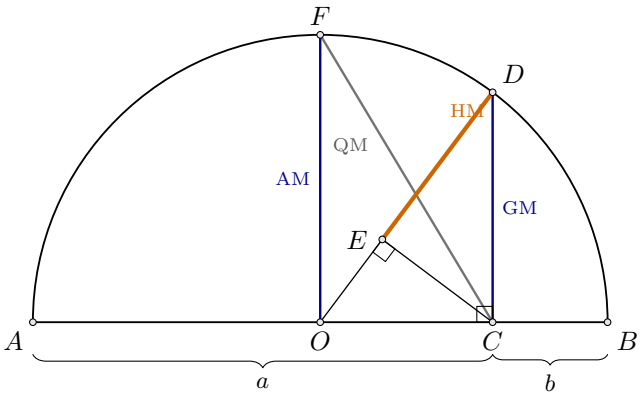
$$\sqrt{\frac{a^2 + b^2}{2}} \geq \frac{a + b}{2} \geq \sqrt{ab} \geq \frac{2ab}{a + b},$$

with equality throughout if and only if $a = b$.

For the first inequality, square both sides: $\frac{a^2+b^2}{2} - \frac{(a+b)^2}{4} = \frac{(a-b)^2}{4} \geq 0$. The middle one is AM-GM. For the last, note that $\frac{2ab}{a+b} = \frac{ab}{(a+b)/2}$, so the ratio of the geometric mean to the harmonic mean equals the ratio of the arithmetic mean to the geometric mean, which is at least 1.

There is also a geometric proof. Take a semicircle on a diameter AB of length $a + b$, with centre O, and mark C on AB with $AC = a$ and $CB = b$. Then the radius OF perpendicular to AB is the arithmetic mean. The perpendicular CD to AB is the geometric mean, since $CD^2 = AC \cdot CB$ by similar triangles. Dropping the perpendicular CE to OD gives $DE \cdot DO = DC^2$, so DE is the harmonic mean. Finally $OC = \frac{|a-b|}{2}$, so Pythagoras gives $CF = \sqrt{\frac{a^2+b^2}{2}}$.

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Each inequality is now "a hypotenuse is longer than a leg": $CF \geq OF$, $OD \geq CD$ and $CD \geq DE$. In contests, QM-AM is the tool when you know a sum of squares and want to bound a sum (or the reverse), and the harmonic mean turns up whenever rates are averaged.

1.5 Extreme cases and bounding

Not every optimisation problem is AM-GM. Two habits cover most of the rest. First, *test the extreme cases before you prove anything*: make the variables equal, make one of them as large as possible, set some to zero. In symmetric problems one end of the range is often at the symmetric point and the other at a boundary. These tests tell you what answer to aim for, and so which inequality you need. Second, *if you bound pieces separately, check that the bounds are reached at the same point*. If you prove $A \leq 3$ and $B \leq 1$, then $A + B \leq 4$, but that is only the maximum if both equalities can hold together.

Example 4.
Non-negative reals a, b, c satisfy $a + b + c = 3$. Find the largest and smallest possible values of $S = ab + bc + ca + abc$.

Solution.
Test some cases first. At $a = b = c = 1$ we get $S = 3 + 1 = 4$. At $(3, 0, 0)$ we get $S = 0$. At $(\frac{3}{2}, \frac{3}{2}, 0)$ we get $S = \frac{9}{4}$. So we guess the maximum is 4 at the symmetric point and the minimum is 0 at the boundary.
The minimum is immediate: every term of S is non-negative, so $S \geq 0$, and $(3, 0, 0)$ gives 0.
For the maximum, bound the two parts separately. Since

$$(a+b+c)^2 - 3(ab+bc+ca) = a^2+b^2+c^2-ab-bc-ca = \tfrac{1}{2}\big((a-b)^2+(b-c)^2+(c-a)^2\big) \geq 0,$$

we get $ab + bc + ca \leq \frac{9}{3} = 3$. By AM-GM, $abc \leq \left(\frac{a+b+c}{3}\right)^3 = 1$. So $S \leq 4$, and both bounds are equalities at $a = b = c = 1$, where $S = 4$. The largest value is 4 and the smallest is 0.
Splitting worked only because both pieces peak at the same point. When they peak at different points, separate bounds give a true statement that is not the answer.

Common traps.

- Equality must be achievable. For $x \geq 5$ it is true that $x + \frac{9}{x} \geq 6$, but equality would need $x = 3$, which is not allowed. The real minimum is $5 + \frac{9}{5} = \frac{34}{5}$, because the expression increases for $x \geq 3$.

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- AM-GM needs non-negative numbers. The claim $x + \frac{1}{x} \geq 2$ is false for negative x.
- A bound that depends on the variable, such as $2\sqrt{54/x}$ in Example 2, is not a bound.
- Separate bounds must be attained at the same point, as in Example 4.
- Check that a maximum really exists. On $x > 0$, the expression $1 - \frac{1}{x}$ is always below 1 and gets as close as you like, but never reaches it, so it has no maximum.

1.6 How to spot this in a contest

The phrase "smallest possible value" or "largest possible value" is the obvious signal. The shape of the expression then suggests the method:

1. A sum of terms whose product is constant, or a product with a fixed sum: AM-GM, after splitting if needed.
2. A quadratic in one or more variables: complete the square.
3. A symmetric constraint: test equal values and boundary values first.
4. Integer variables: usually an extreme-case argument, in which you give the most important variable its best possible value and check that nothing beats it.

On a multiple-choice paper, finding the equality case and evaluating there often identifies the answer quickly. Keep the proof in mind anyway, because the same problems reappear at olympiad level where only the proof scores.

2 Practice problems

The problems are roughly in order of difficulty, from $\star$ to $\star\star\star\star$. Hints are in Section 4 and full solutions in Section 5.

Problem 1 $\star$ For positive x, find the smallest value of $9x + \frac{16}{x}$.

Problem 2 $\star\star$ Real numbers x and y satisfy $x + 2y = 10$. Find the smallest possible value of $x^2 + y^2$.

Problem 3 $\star\star$ Find the largest value of $\frac{x}{x^2+16}$ over all real x.

Problem 4 $\star\star\star$ The equation $x^4 - 12x^3 + ax^2 + bx + 81 = 0$ has four positive real roots, not necessarily distinct. Find $a - b$.

Problem 5 $\star\star\star$ Positive reals a, b and c satisfy $a + b + c = 1$. Find the largest possible value of ab^2c^3.

Problem 6 $\star\star\star\star$ Real numbers x, y and z satisfy $x + y + z = 0$ and $x^2 + y^2 + z^2 = 6$. Find the largest possible value of xyz.

3 Contest ladder

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Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **UKMT SMC 2020 Question 19** ⋆
   Combine the two equations into one expression for t, find its extreme values, then check the values in between
2. **AMC 12A 2021 Problem 7** ⋆⋆
   Expand, then look for a factorisation into a product of two simple pieces
3. **UKMT SMC 2017 Question 18** ⋆⋆
   Clear the fractions, square, and factorise the resulting quadratic in x and y
4. **AHSME 1965 Problem 34** ⋆⋆
   Substitute $u = 1 + x$ and split the fraction into two terms with constant product
5. **UKMT SMC 2021 Question 23** ⋆⋆⋆
   Pair the factors to make a difference of two squares
6. **AHSME 1979 Problem 29** ⋆⋆⋆
   Simplify the fraction with a difference of squares before optimising
7. **AMC 12A 2003 Problem 24** ⋆⋆⋆
   Write everything in terms of the single quantity $\log_a b$
8. **UKMT SMC 2022 Question 24** ⋆⋆⋆
   Make y the subject and divide out, then ask how small a positive divisor can be
9. **AHSME 1982 Problem 29** ⋆⋆⋆⋆
   The minimum of a product under these constraints sits at an extreme case
10. **AMC 12A 2016 Problem 24** ⋆⋆⋆⋆
    Combine Vieta's formulas with AM-GM on the three roots

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4 Hints

1. The product of the two terms does not depend on x.
2. Use the constraint to remove x, then complete the square.
3. Only positive x can give a positive value. For those, divide the top and bottom by x.
4. Use Vieta to find the sum and the product of the roots, then compare their arithmetic and geometric means.
5. Split b into two equal pieces and c into three, so that you have six pieces with a fixed sum.
6. Fix z. Express $x+y$ and xy in terms of z, and ask what it takes for x and y to be real.

5 Solutions

Problem 1

For positive x, find the smallest value of $9x + \frac{16}{x}$.

By AM-GM, $9x + \frac{16}{x} \geq 2\sqrt{9x \cdot \frac{16}{x}} = 2\sqrt{144} = 24$. Equality holds when $9x = \frac{16}{x}$, that is $x = \frac{4}{3}$, and then the expression is $12 + 12 = 24$. So the minimum is 24.

Answer: 24

Problem 2

Real numbers x and y satisfy $x + 2y = 10$. Find the smallest possible value of $x^2 + y^2$.

Substituting $x = 10 - 2y$ gives $x^2 + y^2 = (10-2y)^2 + y^2 = 5y^2 - 40y + 100 = 5(y-4)^2 + 20$. This is at least 20, with equality when $y = 4$ and $x = 2$. So the minimum is 20. Geometrically, $x^2 + y^2$ is the squared distance from the origin to a point of the line $x + 2y = 10$, and the closest point is $(2, 4)$.

Answer: 20

Problem 3

Find the largest value of $\frac{x}{x^2+16}$ over all real x.

For $x \leq 0$ the fraction is not positive, so the maximum, if it exists, comes from $x > 0$. For $x > 0$ we can write $\frac{x}{x^2+16} = \frac{1}{x+\frac{16}{x}}$. By AM-GM, $x + \frac{16}{x} \geq 2\sqrt{16} = 8$, with equality at $x = 4$. So the fraction is at most $\frac{1}{8}$, and it equals $\frac{1}{8}$ at $x = 4$. The largest value is $\frac{1}{8}$.

Answer: $1/8$

Problem 4

The equation $x^4 - 12x^3 + ax^2 + bx + 81 = 0$ has four positive real roots, not necessarily distinct. Find $a - b$.

Let the roots be $r_1, r_2, r_3, r_4 > 0$. By Vieta's formulas, $r_1 + r_2 + r_3 + r_4 = 12$ and $r_1 r_2 r_3 r_4 = 81$. So the arithmetic mean of the roots is 3, and their geometric mean is $\sqrt[4]{81} = 3$ as well. AM-GM says

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that the arithmetic mean is at least the geometric mean, with equality only when all the numbers are equal. Here equality holds, so every root equals 3.

Hence the polynomial is $(x-3)^4 = x^4 - 12x^3 + 54x^2 - 108x + 81$, giving $a = 54$, $b = -108$ and $a - b = 162$.

Answer: 162

Problem 5

Positive reals a, b and c satisfy $a + b + c = 1$. Find the largest possible value of ab^2c^3.

Write $1 = a + \frac{b}{2} + \frac{b}{2} + \frac{c}{3} + \frac{c}{3} + \frac{c}{3}$ and apply AM-GM to these six positive numbers:

$$\frac{1}{6} \geq \sqrt[6]{a \cdot \frac{b^2}{4} \cdot \frac{c^3}{27}}.$$

Raising both sides to the sixth power gives $\frac{ab^2c^3}{108} \leq \frac{1}{6^6}$, so $ab^2c^3 \leq \frac{108}{46656} = \frac{1}{432}$.

Equality needs $a = \frac{b}{2} = \frac{c}{3}$, and together with $a + b + c = 1$ this gives $a = \frac{1}{6}$, $b = \frac{1}{3}$ and $c = \frac{1}{2}$. There $ab^2c^3 = \frac{1}{6} \cdot \frac{1}{9} \cdot \frac{1}{8} = \frac{1}{432}$, so the largest value is $\frac{1}{432}$. The split was chosen so that the six pieces can all be equal at the same time as the constraint holds, which is what makes the bound attainable.

Answer: $1/432$

Problem 6

Real numbers x, y and z satisfy $x + y + z = 0$ and $x^2 + y^2 + z^2 = 6$. Find the largest possible value of xyz.

Squaring $x + y + z = 0$ gives $x^2 + y^2 + z^2 + 2(xy + yz + zx) = 0$, so $xy + yz + zx = -3$. Now fix z. Then $x + y = -z$ and $xy = -3 - z(x + y) = z^2 - 3$. So x and y are the roots of $t^2 + zt + z^2 - 3 = 0$, and they are real exactly when the discriminant is non-negative: $z^2 - 4(z^2 - 3) \geq 0$, that is $z^2 \leq 4$. Conversely, any z with $-2 \leq z \leq 2$ gives real x and y satisfying both conditions.

Now $xyz = z(z^2 - 3) = z^3 - 3z$, and we want its largest value for $-2 \leq z \leq 2$. We have

$$2 - (z^3 - 3z) = -(z - 2)(z + 1)^2 = (2 - z)(z + 1)^2 \geq 0$$

for $z \leq 2$, so $xyz \leq 2$. Equality holds at $z = 2$, where $x = y = -1$, and these values satisfy both conditions, with $xyz = 2$. So the largest value is 2.

Answer: 2